Market Access and the Production of Information

Job Market Paper

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ABSTRACT

Market access changes which assets investors can hold and, in turn, reshapes how they allocate scarce data-processing capacity across assets—including assets they cannot hold. Investors with partial access engage in cross-asset learning only when the auxiliary asset is informative enough to be useful but not so informative that its information is already efficiently revealed through prices. As a result, market access can separate different dimensions of informational efficiency: making an asset investable can reduce its price informativeness, while in other cases prices become more informative even as average posterior precision falls. A China calibration shows that partial market access can generate large equilibrium effects by substantially reshaping information allocation and price informativeness despite a small foreign ownership share. Expanding foreign investability further has little impact, while relaxing the short-sale constraint substantially improves price informativeness.

Keywords: market access; information acquisition; price informativeness; cross-asset learning; financial liberalization

JEL classification: D83, G11, G14, G15

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I. Introduction

Restrictions on which assets investors may hold change what information they find valuable to acquire and process. This paper studies the information channel of market access: when investor groups differ in their information endowments and data-processing technologies, asset-specific access restrictions shape how scarce data-processing capacity is allocated across assets—including assets that some investors cannot hold—and how that information is aggregated in prices. This mechanism applies broadly to settings with segmented investor bases, eligibility rules, or portfolio mandates. A leading application is opening a domestic stock market to foreign investors.

To study this mechanism in a concrete setting, this paper develops a model of stock-market liberalization. Domestic investors alone possess soft information generated by local knowledge and relationships, while hard information can be acquired through data processing by both foreign and domestic institutions, with foreign institutions enjoying better data-processing technology. Thus neither group is uniformly better informed: domestic investors benefit from soft information, whereas foreign institutions have a data-processing advantage. This heterogeneity is embedded in a noisy rational-expectations model with endogenous information acquisition, in which institutions allocate scarce data-processing capacity across assets before trading.

The model shows that investability and information relevance need not coincide. Under partial access, foreign institutions cannot take positions in Type-II shares while their information can nevertheless be valuable for Type-I trading when the two payoffs are informative about one another. The relevant object is therefore not whether an investor can hold an asset, but whether its payoff contains information useful for valuing an asset the investor can hold. This cross-asset channel gives market access a broader effect than a one-asset model would imply: opening one segment can alter data acquisition and prices in another segment that remains closed.

The model yields a sharp condition for this cross-asset learning. Investors with partial access acquire information about the inaccessible asset only at intermediate precisions: information that is too noisy is not worth acquiring, while information that is sufficiently precise is already reflected in equilibrium prices. When they acquire such information, it improves their valuation and demand for the asset they can trade. Their demand therefore transmits information about the inaccessible asset into the equilibrium price signal observed by other investors. Those investors update beliefs, reallocate data-processing capacity, and adjust portfolios, so cross-asset learning affects both the private allocation of information and its aggregation in prices.

As a result, market access can separate different dimensions of informational efficiency. Making an asset investable can reduce its price informativeness. In other cases, market access increases price informativeness while reducing average investor information. A policy that increases the information held by some investors therefore need not improve the information revealed to the market as a whole. These effects arise from the joint response of private information acquisition and price aggregation, rather than from a mechanical relation between liberalization and information.

Portfolio constraints add a separate margin to this mechanism. With a short-sale constraint, information affects whether an investor participates in a market as well as the position taken conditional on participation. Constraints can therefore weaken the incorporation of private information into prices even when investors have strong incentives to acquire that information. Expanding the set of assets available to a group and relaxing its trading constraints are consequently economically distinct interventions.

The model is calibrated to the Shanghai–Hong Kong Stock Connect program. Despite a small foreign ownership share, partial market access can substantially reshape equilibrium information allocation and price informativeness. These effects arise from information transmission rather than from the conventional capital-flow mechanism: opening the foreign-investable segment changes foreign institutions' informed demand for Type-I shares and thereby alters the price signal from which domestic investors learn. Domestic institutions then reallocate data-processing capacity and adjust portfolios across both asset types. Foreign ownership is therefore not a sufficient statistic for equilibrium informational influence.

The counterfactuals distinguish incremental market access from trading restrictions. At the estimated Chinese benchmark, expanding foreign investability from partial to full access has little effect when short-sale constraints remain in place. Relaxing the short-sale constraint, however, substantially improves price aggregation. Portfolio flexibility, rather than further investability alone, is therefore the quantitatively dominant policy margin in this calibration. In the benchmark, foreign institutions are small in both investor mass and observed ownership, and their long positions are infrequent under the short-sale constraint. Their informational role can nevertheless be consequential because information affects marginal price-setting demand and is transmitted through a public price signal to the larger domestic sector.

The structural analysis is complemented with reduced-form evidence around the Stock Connect announcement and launch. Stock Connect created a clear difference between shares that

became foreign-investable and those that remained inaccessible under partial access, and eligible shares exhibit differential movements in return-based information outcomes. These patterns corroborate the central premise that market access can matter for the information environment.

The remainder of the paper proceeds as follows. Section II briefly reviews the related literature. Section III sets the general modeling framework, while Section IV derives the mechanism governing information allocation and aggregation across market-access regimes and Section V introduces portfolio constraints and endogenous participation. Section VI studies welfare. Section VII calibrates the model to Stock Connect, evaluates counterfactual policies, and presents reduced-form evidence. The final section concludes.

II. Literature

Market access determines not only who can trade a security but also what information heterogeneous investors choose to produce. This paper brings this information-allocation implication to the literature on market segmentation and financial liberalization. A large literature studies risk sharing, valuation, and the cost of capital around liberalization episodes (Henry, 2000, 2003; Bekaert and Harvey, 2000; Kim and Singal, 2000; Chari and Henry, 2004), while related research documents changes in domestic information environments, including analyst activity, disclosure, and the incorporation of firm-specific information into prices (Bae et al., 2006; Chan and Kwok, 2017; Griffin et al., 2010). By making data acquisition endogenous, the paper shows that access restrictions govern not only who may hold a security but also where investors direct scarce data-processing capacity. Expanding market access can therefore affect prices through information allocation and aggregation, in addition to the conventional risk-sharing and capital-flow mechanisms.

In light of this information channel, empirical work documents both local information advantages and foreign advantages associated with sophistication and information processing (Choe et al., 2005; Dvořák, 2005; Hau, 2001; Grinblatt and Keloharju, 2000). This paper fills the corresponding gap in the theoretical literature and formalizes this multidimensional heterogeneity by separating locally endowed soft information from universally available hard information that can be acquired through processing data (Liberti and Petersen, 2019; Gao and Zeng, 2023; Gao and Huang, 2020). It studies how asset-specific market access governs the interaction of these two advantages. The distinction is important because local information and processing technology

need not rank investor types in the same way. Domestic institutions may be better placed to interpret local conditions, while foreign institutions may extract higher precision from a common data source.

The theoretical framework builds on noisy rational-expectations models of price aggregation (Grossman and Stiglitz, 1980; Hellwig, 1980; Diamond and Verrecchia, 1981; Verrecchia, 1982), which study how prices aggregate dispersed private information, and on endogenous-information-acquisition models (Van Nieuwerburgh and Veldkamp, 2009), which determine investors' information choices jointly with their portfolio decisions. In these frameworks, the assets investors may trade are generally taken as given. This paper instead makes asset-specific market access an input into that joint problem, so that information about an inaccessible asset can improve beliefs about an accessible one. The resulting cross-asset learning differs from conventional diversification: investors acquire information about an inaccessible asset not to hold it directly, but because it improves the valuation and trading of an accessible asset.

III. General Framework

This section formalizes the general mechanism of asset-specific access across heterogeneous investor groups using the domestic–foreign structure and provides the common framework on which the different access regimes build. One group represents locally informed domestic investors, and the other represents investors with superior data-processing technology.

A. Assets

There is a domestic safe asset with the rate of return r. Domestic firms issue two aggregate risky securities: a type-I security that is eligible for international trade under partial access, and a type-II risky asset that remains closed to international investors under partial access and becomes investable under full access. Let $\boldsymbol{d}$ denote the dividend state known at the start of the trading date. The end-of-date dividend $\widetilde{\boldsymbol{d}}$ follows $\widetilde{\boldsymbol{d}} = \boldsymbol{\mu} + \boldsymbol{G}\boldsymbol{d} + \widetilde{\boldsymbol{y}}$, where $\boldsymbol{G} \equiv \text{diag}(g_1, g_2)$ is the dividend-growth matrix and $\widetilde{\boldsymbol{y}} \overset{i.i.d.}{\sim} N(0, \boldsymbol{\Sigma}_y)$ is the dividend shock. We maintain $r > \max\{g_1, g_2\}$, so discounted dividend values are finite and the asset-specific mappings from effective information into absolute signal-to-noise ratios are positive. Write $\boldsymbol{\Omega}_y \equiv \boldsymbol{\Sigma}_y^{-1}$ and let $\mathcal{C}_y \equiv (\boldsymbol{\Omega}_y)_{12}$ denote its off-diagonal cross-precision. In the two-asset Gaussian case, positive fundamental covariance

corresponds to $\mathcal{C}_y < 0$. The dividend is paid out at the end of the trading date. Denote the ex-dividend continuation value at the end of the trading date by $\widetilde{\boldsymbol{v}}$. Then $\widetilde{\boldsymbol{v}} = \boldsymbol{\mu}_v + (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}\widetilde{\boldsymbol{d}}$, where $\boldsymbol{\mu}_v = \frac{r}{r-1}(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu}$.

B. Investors

There is a continuum of investors with four potential types: domestic institutional investors, domestic individual investors, foreign institutional investors, and foreign individual investors, with respective masses λ_I, λ_J, λ_K, and λ_H. Which types are present is regime-specific: let $\mathcal{L}^R \subseteq \mathcal{L} \equiv \{I, J, K, H\}$ denote the types active at the trading date under regime R; the regime-specific active sets used below are stated at the end of this section. For simplicity, the following sections use $l \in \{i, j, k, h\}$ to refer to an investor of any type, and $L \in \{I, J, K, H\}$ to refer to an investor type. Investors of type L are endowed with initial wealth w_L. Which types are active is determined by the access regime alone: the model is a single trading-date economy, and there is no entry or exit across dates. Individual investors do not acquire or process data, while institutional investors acquire and process data, obtaining private signals about the dividend innovation. All domestic investors are additionally endowed with soft information about the dividend innovation.

All investors have CARA utility over end-of-date wealth, with risk-aversion coefficient ρ, so $U_l = -\exp(-\rho\widetilde{w}_l')$, $l \in \{i, j, k, h\}$. Investors make portfolio choices $\boldsymbol{q}_l$ to maximize expected utility conditional on their respective information sets and subject to a one-period budget constraint

$$w_l' = rw_l + \boldsymbol{q}_l'(\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}r) \tag{1}$$

The initial wealth w_K of foreign investors is entirely invested in domestic securities markets; their wealth invested in foreign economies is not modeled.

C. Market access and portfolio constraints

The economy is a single trading-date model. We compare equilibria across access regimes $R \in \{\text{cl}, \text{pa}, \text{fu}\}$ (closed, partial, and full access) and across portfolio regimes, with or without a short-sale constraint, at the same trading date rather than across a sequence of periods. Partial access without a short-sale constraint is unmarked.

Domestic investors I and J can trade both types of risky assets in every access regime, with

feasible set $\mathbb{R}^2_+$ and $\mathbb{R}^2$ with and without a short-sale constraint, respectively. Foreign investors are absent under closed access. Under partial access, foreign institutions K can trade only the Type-I asset, so their feasible set is $\mathbb{R} \times \{0\}$ or $\mathbb{R}_+ \times \{0\}$, while foreign individuals are still absent. Under full access, foreign institutions and individuals can trade both assets, with feasible sets $\mathbb{R}^2$ or $\mathbb{R}^2_+$.

D. Information structure

Before making portfolio choices at the trading date, institutional investors can acquire data to generate private signals about the end-of-date dividend innovation $\widetilde{\boldsymbol{y}}$. In addition, domestic investors are endowed with soft information[1] about $\widetilde{\boldsymbol{y}}$. The equilibrium price aggregates all private information.

Investors have different information sets at different stages of the decision process. When making data-acquisition decisions, domestic institutional investor i has only the prior and soft information $\boldsymbol{\xi}_i$, denoted $\mathcal{I}_{i^-} = \{\boldsymbol{\xi}_i\}$, while foreign institutional investor k, when active, has no relevant information beyond the prior. Let $\boldsymbol{s}_l$, $l \in \{i, k\}$, denote the private signals generated from data, and let $\boldsymbol{s}_p$ denote the price signal. When investor i selects its portfolio, the choice is based on the equilibrium price, soft information, and the private signal generated from acquired data, so $\mathcal{I}_i = \{\boldsymbol{s}_p, \boldsymbol{s}_i, \boldsymbol{\xi}_i\}$. Domestic individual investor j does not acquire or process data and bases its portfolio choice only on endowed soft information and the equilibrium price, so $\mathcal{I}_j = \{\boldsymbol{s}_p, \boldsymbol{\xi}_j\}$. Foreign individual investors H, when active, have no soft information and base their portfolio choice only on the equilibrium price, so $\mathcal{I}_h = \{\boldsymbol{s}_p\}$.

The general equilibrium definition below does not restrict the price mapping. In the unrestricted economies of Section IV, the equilibrium price is a linear function of the signals, and since signals are normally distributed, the price can be expressed as a linear function of shocks:

$$\boldsymbol{p} = \boldsymbol{A} + \boldsymbol{B}\boldsymbol{d} + \boldsymbol{C}\widetilde{\boldsymbol{y}} + \boldsymbol{D}\widetilde{\boldsymbol{x}}. \tag{2}$$

When $\boldsymbol{D}$ is nonsingular, define the signal-to-noise ratio (SNR) as $\boldsymbol{\zeta} \equiv \boldsymbol{D}^{-1}\boldsymbol{C}$. An affine transformation of the equilibrium price $\boldsymbol{p}$ generates an unbiased signal about the end-of-date dividend

[1] Domestic institutional investors may plausibly possess more soft information than domestic individual investors. Because this distinction is not central to the paper and does not alter the model's structure or results, the model assumes that all domestic investors have the same soft-information precision.

shock. At the trading date, investors extract the signal $\boldsymbol{s}_p = \widetilde{\boldsymbol{y}} + \boldsymbol{\zeta}^{-1}\widetilde{\boldsymbol{x}}$, whose precision is $\boldsymbol{\Omega}_p = \boldsymbol{\zeta}'\boldsymbol{\Omega}_x\boldsymbol{\zeta}$. Under the short-sale regimes of Section V, exact constrained aggregation makes the global price mapping nonlinear, and that section characterizes its local linear response around a reference state using the same price-signal representation.

Throughout the paper, we distinguish data from information. Data are purchasable inputs, such as accounting information, prices and trading volume, or textual disclosures. Information is the private signal about the unknown dividend innovation $\widetilde{\boldsymbol{y}}$ that results from processing those data. Investors may differ in their data sources, acquired quantities, and data-processing technologies.

Specifically in the model, institutional investors can acquire a total of $\bar{M}$ data points from two data sources. The data acquisition costs are normalized to zero. Thus investors acquire data at full capacity $\bar{M}$. Denote the numbers of data points acquired by investor l at the trading date for type-I and type-II risky assets as $M_{l,(\mathrm{I})}$ and $M_{l,(\mathrm{II})}$, respectively. Each data source corresponds to one type of firm. The model represents processing technology directly through the quality of the resulting unbiased hard-information signal. A foreign institution that processes a Type-N observation obtains

$$s^{(m)}_{K,(N)} = \widetilde{y}_{(N)} + u^{(m)}_{K,(N)}, \qquad u^{(m)}_{K,(N)} \sim N(0, \Pi^{-1}_{e,(N)}),$$

whereas a domestic institution obtains

$$s^{(m)}_{I,(N)} = \widetilde{y}_{(N)} + u^{(m)}_{I,(N)}, \qquad u^{(m)}_{I,(N)} \sim N\big(0, (\varphi^2\Pi_{e,(N)})^{-1}\big)\,.$$

Signal noises are independent across observations, investors, and data sources, and $\boldsymbol{\Pi}_e$ is diagonal. Thus one foreign-processed observation contributes $\Pi_{e,(N)}$ to posterior precision, while one domestically processed observation contributes $\varphi^2\Pi_{e,(N)}$. For continuity with the technology notation used below, write $\boldsymbol{\Pi}_e \equiv \beta^2\boldsymbol{\Omega}_e$ and $\varphi \equiv \beta_I/\beta \in (0,1)$, where β/β_I indexes the foreign–domestic processing advantage. This is a reduced-form signal-quality wedge: it does not represent domestic institutions as applying an incorrect loading to the same raw observation. The economically operative objects are therefore $(\boldsymbol{\Pi}_e, \varphi)$.

The resulting signal is *hard information*. By contrast, *soft information* cannot be stored or purchased as data and is held only by domestic investors. For each domestic investor $l \in \{i, j\}$,

let

$$\widetilde{\boldsymbol{\xi}}_l = \widetilde{\boldsymbol{y}} + \widetilde{\boldsymbol{\epsilon}}_l, \qquad \widetilde{\boldsymbol{\epsilon}}_l \overset{i.i.d.}{\sim} N(0, \boldsymbol{\Sigma}_\epsilon).$$

Although domestic institutional investors have an informational advantage with the soft information, they have a technological disadvantage over processing data and are not able to implement the perfect data processing technology. Thus a priori it is unclear who has an overall information advantage.

D.1. Equilibrium

The regime-specific structure can be summarized by two objects: the set of active types $\mathcal{L}^R$ and the feasible portfolio correspondence $\mathcal{Q}_L^R \subseteq \mathbb{R}^2$. At the trading date, $\mathcal{L}^R = \{I, J\}$ if market access is denied to foreign investors, $\mathcal{L}^R = \{I, J, K\}$ if partial access is granted to foreign institutional investors, and $\mathcal{L}^R = \{I, J, K, H\}$ if full access is granted to all foreign investors. Partial access imposes $\mathcal{Q}_K^{\text{pa}} = \mathbb{R} \times \{0\}$, whereas full access has $\mathcal{Q}_K^{\text{fu}} = \mathbb{R}^2$; with a short-sale constraint these become $\mathbb{R}_+ \times \{0\}$ and $\mathbb{R}_+^2$, respectively, and every unrestricted feasible set $\mathbb{R}^2$ is replaced by its nonnegative counterpart $\mathbb{R}_+^2$. For an institutional investor, the feasible allocation of data-processing capacity is $\mathcal{M} \equiv \left\{(M_{(\text{I})}, M_{(\text{II})}) \in \mathbb{R}_+^2 : M_{(\text{I})} + M_{(\text{II})} = \bar{M}\right\}$.

Definition 1 (Market equilibrium). *Fix a market regime R. A rational-expectations equilibrium consists of a price function $\boldsymbol{P}^R(\boldsymbol{d}, \widetilde{\boldsymbol{y}}, \widetilde{\boldsymbol{x}})$, data-acquisition strategies for all institutional investors active in regime R, and portfolio strategies $\{\boldsymbol{q}_l^R\}_{l \in \mathcal{L}^R}$, such that:*

- *(i) Each institutional investor chooses its allocation of data-processing capacity from $\mathcal{M}$ to maximize expected utility at the trading stage conditional on its information at the data-choice stage, taking the equilibrium price-information environment and the subsequent optimal portfolio choice as given;*
- *(ii) For every investor l of type $L \in \mathcal{L}^R$, $\boldsymbol{q}_l^R \in \arg\max_{\boldsymbol{q} \in \mathcal{Q}_L^R} \mathbb{E}\left[-\exp(-\rho \widetilde{w}_l') \mid \mathcal{I}_l\right]$, subject to the budget constraint* (1)*;*
- *(iii) Investors' conditional beliefs are formed from their private information, soft information when available, and the information contained in equilibrium prices;*
- *(iv) The price function clears the market: the aggregate investor holdings of the risky assets equal the exogenous net position $\bar{\boldsymbol{x}} + \widetilde{\boldsymbol{x}}$.*

IV. Market Access and Endogenous Information Allocation

Building on the modeling framework, this section studies how asset-specific access shapes endogenous information allocation before portfolio constraints are introduced. An access-restricted group may acquire information about an asset it cannot hold when that information improves trading of an accessible asset. The closed- and full-access regimes then show how relaxing or removing this restriction changes the allocation of fixed processing capacity and the aggregation of information in prices. Specifically using the domestic–foreign language, domestic institutions have access to both assets, whereas foreign institutions are restricted to Type-I shares.

A. Partial access

The modeling framework is specified to the partial-access regime where $\mathcal{L} = \{I, J, K\}$, $\mathcal{Q}_I = \mathcal{Q}_J = \mathbb{R}^2$ and $\mathcal{Q}_K = \mathbb{R} \times \{0\}$.

A.1. Portfolio choices

At the trading stage, investor l faces a simple one-shot portfolio choice problem, which has an optimal solution[2]

$$\boldsymbol{q}_l = \frac{1}{\rho}\hat{\boldsymbol{\Omega}}_L \hat{\boldsymbol{\mu}}_l, \tag{3}$$

where $\hat{\boldsymbol{\mu}}_l \equiv \mathbb{E}[\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}r \mid \mathcal{I}_l]$ and $\hat{\boldsymbol{\Omega}}_L \equiv \mathbb{V}ar^{-1}(\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}r \mid \mathcal{I}_l)$ are the conditional mean and precision of excess payoffs.

A.2. Equilibrium price relation

Definition 2 (Cross-precision). *Let $z \equiv \hat{\mathcal{C}}_{Ky}/\hat{\Omega}_{Ky,(\mathrm{II})}$ denote the cross-precision of the foreign posterior, the off-diagonal entry of $\hat{\boldsymbol{\Omega}}_{Ky}$ normalized by its $\{2,2\}$ element.*

Economically, z measures the intensity of foreign *cross-asset learning*: although foreign institutions can trade only Type-I shares, information about Type II can affect their Type-I demand when fundamentals are correlated.

[2]For notational simplicity, this paper adopts the notational style that uses a hat to distinguish conditional moments at the portfolio choice stage (*interim*) with those at the data choice stage (ex ante).

Signal-to-noise ratio The SNR enters the market clearing condition and is related to other variables through

$$\boldsymbol{\zeta} = -\frac{1}{\rho r}(r\boldsymbol{I} - \boldsymbol{G})\Big[\Big(\lambda_I \varphi^2 \boldsymbol{M}_I + \lambda_K \Big(\boldsymbol{E}_{11} - z\boldsymbol{E}_{12}\Big)\boldsymbol{M}_K\Big)\boldsymbol{\Pi}_e + \lambda_1 \boldsymbol{\Omega}_\epsilon\Big], \tag{4}$$

where $\lambda_1 = \lambda_I + \lambda_J$, $\mathbb{E}_{11} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $\mathbb{E}_{12} = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. An interesting observation is that the SNR matrix is not symmetric. If foreign institutional investors choose to acquire data on type-Ⅱ risky asset, the information they obtain after data processing is not directly aggregated into the equilibrium price of type-Ⅱ risky asset since it is not eligible for international trade. However, this information will be partially reflected in the equilibrium price of type-I risky asset, if the two types of assets are correlated, in which case information on type-Ⅱ asset also affects their posterior belief on type-I. This creates informational inefficiency.

Average posterior precision Market clearing aggregates investors' posterior precision into an eligibility-weighted object $\bar{\hat{\boldsymbol{\Omega}}}_y$. Under partial access it need not be symmetric, because it is an equilibrium aggregate rather than an individual Bayesian precision matrix. Foreign entry dilutes type-I soft information because the mass of investors eligible to hold that asset rises from λ_1 to λ_2; it does not mechanically dilute type-Ⅱ soft information, because type-Ⅱ remains closed to foreign investors. Foreign institutions add private information directly when they process type-I data. When the assets are correlated, their type-Ⅱ data can also enter the type-I price through the cross-asset SNR, thereby changing the common price-information component $\boldsymbol{\Omega}_p$ and, in particular, potentially raising $\hat{\Omega}_{p,(\mathrm{II})}$ through the cross-price signal. Limited investability creates two countervailing forces. Foreign Type-II information can improve the information embedded in prices through cross-asset learning, but the trading restriction limits the precision that foreign institutions can use directly in Type-I demand.

Price relation Let $\boldsymbol{\Delta}_{\text{pa}}$ denote the eligibility correction induced by foreign Type-II ineligibility; it vanishes under full access. In equilibrium, the price coefficients are characterized by

$$\boldsymbol{A} = \frac{1}{r}\boldsymbol{\mu}_v + (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu} - \rho r(r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y\boldsymbol{\Lambda}^{-1}(r\boldsymbol{I} - \boldsymbol{G})^{-1}\bar{\boldsymbol{x}}, \tag{5}$$

$$\boldsymbol{B} = (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}, \tag{6}$$

$$\boldsymbol{C} = (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\left[\boldsymbol{I} - \bar{\hat{\boldsymbol{\Sigma}}}_y\boldsymbol{\Omega}_y + \bar{\hat{\boldsymbol{\Sigma}}}_y\boldsymbol{\Delta}_{\text{pa}}\right], \tag{7}$$

$$\boldsymbol{D} = \boldsymbol{C}\boldsymbol{\zeta}^{-1}, \tag{8}$$

where $\bar{\hat{\boldsymbol{\Sigma}}}_y \equiv \bar{\hat{\boldsymbol{\Omega}}}_y^{-1}$. $\boldsymbol{B}$ is the constant discounted dividend growth rate, and together $\boldsymbol{A}$ and $\boldsymbol{B}$ govern the level of the equilibrium price. $\boldsymbol{C}$ is the price sensitivity to dividend innovation, and $\boldsymbol{D}$ regulates the amount of demand shocks contained in the price. Purchased data amount $\boldsymbol{M}_L$, data-processing capacity $\bar{M}$, effective data precision $\boldsymbol{\Pi}_e$, relative data-processing efficiency φ, and the precision of soft information $\boldsymbol{\Omega}_\epsilon$ all enter the price coefficients via the SNR and the average posterior precision of the dividend innovation.

A.3. Endogenous information allocation

The central equilibrium choice is how each institutional investor allocates the fixed capacity $\bar{M}$ across the two data sources. More generally, investor groups face different allocation problems when they differ in endowed information and in the assets to which they have access. In the implementation, domestic investors possess soft information while foreign institutions are granted access only to the Type-I asset. The two investor problems are characterized in turn.

Domestic information allocation The relevant sufficient statistic for the allocation problem is summarized in the following lemma.

LEMMA 1. *The data choice of domestic institutional investor i at the trading date enters the ex ante utility function only via* $\det(\hat{\boldsymbol{\Sigma}}_i)$*, where $\hat{\boldsymbol{\Sigma}}_i$ is the posterior variance of excess payoffs.*

Since the data choice $(M_{i,(\text{I})}, M_{i,(\text{II})})$ only enter the ex ante utility function through the determinant of the posterior variance of payoff $\det(\hat{\boldsymbol{\Sigma}}_i)$, the data choice problem reduces to allocating data points to maximize the determinant of the posterior precision $\det(\hat{\boldsymbol{\Omega}}_i)$. The F.O.C. can be written in effective-precision units as $\frac{\hat{\Omega}_{iy,(\text{I})}}{\varphi^2\Pi_{e,(\text{I})}} = \frac{\hat{\Omega}_{iy,(\text{II})}}{\varphi^2\Pi_{e,(\text{II})}}$. Thus domestic institutional investor i

equalizes posterior precision measured in units of the effective precision generated by one processed domestic data point. This normalization motivates the following definition.

Definition 3 (Equivalent data amount). *For asset type $N \in \{\mathrm{I}, \mathrm{II}\}$ and access regime a, the equivalent data amount is* $m_{e,(N);a}(M) \equiv (\Omega_{y,(N)} + \hat{\Omega}_{p,(N);a} + \Omega_{\epsilon,(N)})/(\varphi^2 \Pi_{e,(N)}) + M$.

It is the posterior precision for Type-N dividends, expressed in units of the effective precision from one processed domestic data point. Under partial access we suppress the regime tag and write $m_{e,(N)}$. Conditional on the equilibrium price signal, $m_{e,(N);a}(M) = m_{e,(N);a}(0) + M$, so that one additional processed data point raises the corresponding equivalent-data amount by one. This is an individual-choice identity: an atomistic investor takes $\hat{\Omega}_{p,(N);a}$ as given when reallocating processing capacity. Across equilibrium regimes, price precision is endogenous, so the level of the equivalent-data amount can change even holding M fixed. The interior first-order condition is therefore $m_{e,(\mathrm{I})}(M_{i,(\mathrm{I})}) = m_{e,(\mathrm{II})}(M_{i,(\mathrm{II})})$.

Denote the optimal Type-I data allocation of domestic institutional investor i by M_i^*; the corresponding Type-II allocation is $\bar{M} - M_i^*$. The optimal Type-I allocation is

$$M_i^* = \begin{cases} 0, & \text{if } m_{e,(\mathrm{I})}(0) \geq m_{e,(\mathrm{II})}(\bar{M}), \\ \bar{M}, & \text{if } m_{e,(\mathrm{II})}(0) \geq m_{e,(\mathrm{I})}(\bar{M}), \\ \frac{1}{2}\big[m_{e,(\mathrm{II})}(\bar{M}) - m_{e,(\mathrm{I})}(0)\big], & \text{otherwise.} \end{cases} \tag{9}$$

The two corner conditions compare the equivalent-data amounts at the opposite endpoints of the capacity constraint; otherwise, the unique interior allocation equalizes the two equivalent-data amounts. Thus the domestic allocation follows a water-filling rule: more processing capacity is directed to the asset with the lower initial equivalent-data stock.

Foreign cross-asset learning Under partial access, foreign institutions can trade Type I but cannot trade Type II. Their information problem is nevertheless two-dimensional: because fundamentals are correlated, processing Type-II data can improve the posterior used to trade Type I. Thus information can have private trading value even when the underlying asset is not directly tradable. This is the general cross-asset-learning channel: an access-restricted group may optimally acquire information about an inaccessible asset when that information improves decisions in an accessible market. The following lemma identifies the sufficient statistic for this information choice in the domestic–foreign implementation.

LEMMA 2 (Foreign data choice under partial access). *Conditional on the equilibrium price signal, a foreign institution allocates data to maximize the Type-I Schur-complement precision*

$$\hat{\Omega}^{S}_{Ky,(\mathrm{I})} = \hat{\Omega}_{Ky,(\mathrm{I})} - \frac{\hat{\mathcal{C}}^{2}_{Ky}}{\hat{\Omega}_{Ky,(\mathrm{II})}}.$$

Thus the Type-II trading restriction does not make Type-II information irrelevant: Type-II data can raise the effective precision used in the foreign investor's Type-I trading problem through the Schur complement. To proceed, define the marginal value of this cross-asset learning.

DEFINITION 4 (Information spillover rate). *Conditional on the equilibrium price signal, let* $M \equiv M_{k,(\mathrm{II})}$ *and write*

$$z(M) \equiv \frac{\mathcal{C}_y + \hat{\mathcal{C}}_p}{\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})} + M\Pi_{e,(\mathrm{II})}}.$$

The information spillover rate is

$$\kappa(M) \equiv z(M)^2 \frac{\Pi_{e,(\mathrm{II})}}{\Pi_{e,(\mathrm{I})}}.$$

It is the marginal Type-I precision generated indirectly by one processed Type-II observation, relative to that generated directly by one processed Type-I observation. The optimal data allocation profile therefore depends on the information spillover rate. If $\kappa(0) \leq 1$, the marginal indirect Type-I precision from Type-II processing is already no greater than the direct Type-I precision at the Type-I corner, so the foreign investor allocates no capacity to Type II, $M^{*}_{k,(\mathrm{II})} = 0$. At the opposite boundary, if $\kappa(\bar{M}) \geq 1$, the indirect marginal precision remains at least as large even after all capacity is assigned to Type II, so $M^{*}_{k,(\mathrm{II})} = \bar{M}$. Otherwise the unique interior allocation equates the two marginal precision gains. The data choice can be expressed as $(\bar{M} - M^{*}_{k,(\mathrm{II})}, M^{*}_{k,(\mathrm{II})})$, where

$$M^{*}_{k,(\mathrm{II})} = \begin{cases} 0, & \kappa(0) \leq 1, \\ \bar{M}, & \kappa(\bar{M}) \geq 1, \\ M^{\text{int}}, & \kappa(0) > 1 > \kappa(\bar{M}). \end{cases} \tag{10}$$

Here, $M^{\text{int}} \in (0, \bar{M})$ is the unique solution to $\kappa(M^{\text{int}}) = 1$. At an interior allocation, this condition means that the marginal value of direct Type-I learning equals the marginal value of indirect Type-I learning through Type-II information.

Conditional on the equilibrium price signal, each institution allocates its processing capacity to equalize marginal information value across the available data sources, subject to its portfolio

constraints.

Proposition 1 (Cross-asset learning under partial access). *Under partial access, a foreign institution that can trade only Type-I shares may nevertheless allocate processing capacity to Type-II data. It does so when the marginal Type-I precision generated indirectly by Type-II processing exceeds that generated by direct Type-I processing. At an interior allocation,*

$$\kappa(M^*_{k,(\mathrm{II})}) = 1,$$

so the marginal gains from direct Type-I learning and cross-asset learning from Type-II information are equal. Hence, investability and information relevance need not coincide.

While foreign institutional investor k could optimally choose to learn about type-II risky asset, the private value of doing so need not coincide with its social value. Whether this choice constitutes excessive information production is determined by the constrained-planner comparison in Section VI; it cannot be inferred from the foreign investor's private problem alone. One implication that does follow directly from the private problem is that cross-asset learning requires fundamental dependence across the two asset types. When the Type-I and Type-II dividend innovations are uncorrelated, learning about Type-II does not improve the foreign investor's belief about the tradable Type-I asset.

Cross-asset learning requires fundamental dependence: an access-restricted group acquires information about an inaccessible asset only when that information improves the valuation of an accessible asset. In the domestic–foreign implementation, this condition is satisfied when Type-I and Type-II fundamentals are correlated. To characterize when this learning occurs, the following comparative static varies effective Type-II precision as an equilibrium object, allowing both information choices and the price signal to adjust jointly.

Proposition 2 (Intermediate-precision cross-asset learning). *Maintain $\mathcal{C}_y < 0$. Let $E \equiv \Pi_{e,(\mathrm{II})}$ vary with primitive Type-II raw-data precision, holding β and all other primitives fixed. Then there exist $0 < \underline{E} < \overline{E} < \infty$ such that*

$$M^*_{k,(\mathrm{II})}(E) = 0$$

for all $E \in (0, \underline{E}) \cup (\overline{E}, \infty)$.

Under the sufficient condition for interior cross-asset learning stated in the Internet Appendix,

there exist $0 < E_- < E_+ < \infty$ *such that*

$$M^*_{k,(\mathrm{II})}(E) > 0$$

for every $E \in (E_-, E_+)$.

At low Type-II precision, indirect learning is unproductive. At high precision, the relevant information is increasingly incorporated into prices and therefore has less private value. These forces generate the intermediate-precision region. Formal bounds, sufficient conditions, and numerical illustrations are provided in the Internet Appendix.

B. Effects of Partial Access

Market access affects both price informativeness and expected returns through its effects on information allocation, price aggregation, and risk bearing. These effects arise from the interaction between heterogeneity in information and processing technology and restrictions on the assets for which investors can deploy their processing capacity. In the domestic–foreign implementation, foreign entry changes the composition of informed demand in the Type-I market and, therefore, the equilibrium price signal observed by domestic investors, who then reallocate data-processing capacity across the two asset types.

The resulting changes in the price signal and in private data acquisition determine the asset-specific average posterior precision, summarized by the diagonal entries $\bar{\tilde{\Omega}}_{y,(N)}$ of the eligibility-weighted precision aggregate $\bar{\tilde{\mathbf{\Omega}}}_y$. Access can therefore separate price informativeness from average posterior precision.

Market access also changes risk bearing. By enlarging the set of investors able to hold eligible shares, foreign entry changes the risk premium embedded in prices, consistent with the risk-sharing component of liberalization emphasized in the literature (Chari and Henry, 2004; Chan and Kwok, 2017).

B.1. Domestic information reallocation

The closed-access counterfactual is readily available from the partial-access model by setting $\lambda_K = 0$. Using Definition 3, $m_{e,(N)}$ without a regime tag denotes the equivalent data amount

evaluated in the partially liberalized equilibrium, while $m_{e,(N);\mathrm{cl}}$ denotes the same normalization evaluated in the closed-access equilibrium ($\lambda_K = 0$). Accordingly, $\hat{\Omega}_{p,(N);\mathrm{cl}}$ is the closed-equilibrium price-signal precision with all foreign-investor terms set to zero. These are regime evaluations of a single concept rather than separate definitions; their levels can differ because equilibrium price precision changes with market access.

B.2. Price informativeness and the signal-to-noise ratio

Partial market access affects the SNRs of type-I and type-II risky assets. Beyond the common channel through changes in domestic institutional investors' data choices, which affects both asset types, the effects differ across assets.

For type-I risky assets, foreign entry can affect the SNR directly or indirectly. If foreign institutions learn about type-I assets, the information extracted from the acquired and processed data is incorporated into the Type-I equilibrium price, directly affecting the Type-I SNR. The effect increases with the foreign investor mass, processing capacity, and data-signal precision. If foreign institutions instead learn about type-II assets, the resulting information is not incorporated directly into the Type-II equilibrium price; it affects the Type-I price only through information spillovers and therefore affects the Type-I SNR indirectly.

For type-II risky assets, domestic institutional investors are the sole contributors to the SNR. Since market access does not affect their endowed soft information, the direction of the SNR change depends only on domestic institutions' data choices: if they learn less about type-II assets, i.e. $M_{(\mathrm{II})} < M_{(\mathrm{II});\mathrm{cl}}$, the Type-II SNR falls; if they learn more, it rises.

Let us call $\zeta_{(\mathrm{I})}$ and $\zeta_{(\mathrm{II})}$ the own-asset SNRs, which measure the loading of each asset's own dividend-innovation signal relative to noise in its equilibrium price, and $\zeta_{(\mathrm{I}\leftarrow\mathrm{II})}$ the cross-asset SNR, which measures the loading of type-II dividend information into the type-I price relative to type-I noise. These ratios depend on the risk-free rate, dividend growth, risk aversion, investor masses, effective data precision $\mathbf{\Pi}_e$, relative processing efficiency φ, and institutional data choices.

Notice that $\zeta_{(\mathrm{I}\leftarrow\mathrm{II})}$ is present only if foreign institutional investors choose to learn about type-II assets. Under the type-II corner $\kappa(\bar{M}) \geq 1$, foreign institutions allocate all processing capacity to type-II data. Their information therefore contributes to the cross-asset SNR rather than directly to either own-asset SNR. Any change in the two own-asset SNRs is then driven by domestic institutions' reallocation of their fixed processing capacity across the two asset types.

The Type-I response reflects two distinct frictions. The positive channel is driven primarily by investability: foreign hard information is aggregated into the Type-I price, with its strength governed by foreign presence and processing technology relative to domestic institutions. A negative response requires the reduction in domestic Type-I learning to be sufficiently strong; the dilution of domestic soft information enters this threshold.

At this all-Type-II foreign-processing corner, the changes normalized by the asset-specific mapping from effective data precision into price informativeness are exactly equal and opposite:

$$\frac{\Delta\left|\zeta_{(\mathrm{I})}\right|}{(r-g_1)\Pi_{e,(\mathrm{I})}} = -\frac{\Delta\left|\zeta_{(\mathrm{II})}\right|}{(r-g_2)\Pi_{e,(\mathrm{II})}}. \tag{11}$$

Thus, at this corner, if domestic institutions increase type-I processing after partial access, the type-I own-asset SNR rises and the type-II own-asset SNR falls; if they decrease type-I processing, the signs reverse. If domestic institutions do not reallocate processing capacity, both own-asset SNR changes are zero. The raw own-asset SNR changes need not be equal in magnitude because the two asset types can differ in effective data precision and dividend-growth exposure.

The opposite-sign movement comes directly from domestic institutions' fixed processing-capacity constraint: more processing of one asset mechanically means less processing of the other. The investability restriction instead explains why, at the Type-II foreign-processing corner, foreign Type-II information enters price informativeness through the cross-asset SNR rather than directly through either own-asset SNR. Equation (11) removes the asset-specific scaling by $(r-g_N)\Pi_{e,(N)}$ and makes the one-for-one domestic reallocation transparent.

The comparison emphasizes that making type-II data investable does not mechanically increase its price informativeness. Foreign institutional investors bring additional effective precision, but this gain can be offset when entry crowds out domestic type-II data acquisition. The exact condition is therefore a comparison between lost domestic effective information and newly created foreign effective information rather than a threshold on foreign investor mass alone.

B.3. Total information and posterior precision

Foreign entry changes total information through three channels: it adds foreign hard information, changes domestic data acquisition, and changes the density of domestic soft information among eligible investors. The last channel mechanically affects Type I but not Type II, because foreign investors become eligible only for Type-I shares under partial access.

The net effect on posterior precision is therefore not determined by price informativeness alone. At the all-Type-I foreign-processing corner, a weak decline in Type-I price informativeness implies a strict decline in average Type-I posterior precision because foreign entry can both reduce domestic information density and impose the portfolio-constraint discount. For Type II, the eligible-investor mass is unchanged, and the cross-price signal can offset a decline in domestic Type-II data acquisition.

C. Closed access

The closed-access regime provides the benchmark in which the second investor group is excluded from the two risky-asset markets. In the domestic–foreign implementation, the domestic financial market remains closed to foreign investors at the trading date, so that only domestic investors trade the two risky securities. The environment is the specialization of Section III in which foreign investor masses are set to zero while the domestic primitives are held fixed. This makes closed access the natural theoretical benchmark for isolating the effect of foreign entry and asset eligibility.

The information problem of the closed economy is the domestic part of the Environment and Market Access: domestic institutional investor i chooses $(M_{i,(\mathrm{I})}, M_{i,(\mathrm{II})})$ subject to the data-processing capacity $\bar{M}$ before observing prices, domestic individual investors j trade on price and soft information only, and the equilibrium price aggregates the private information of domestic institutional investors. With no foreign investors, there is no foreign demand, no foreign data processing, and no dilution of the domestic soft-information density; the price signal $\boldsymbol{s}_p$ is generated by domestic investors alone. Formally, the closed equilibrium specializes Definition 1 to $\mathcal{L}_2 = \{I, J\}$ (equivalently, the $\lambda_K = 0$ case): the equilibrium price function, data strategies, and portfolio choices satisfy the same conditions with all foreign-investor terms set to zero.

D. Full access

Under full access, all domestic risky securities are available to foreign investors. More generally, this regime removes the asset-specific restriction on the second investor group. It is useful both as a theoretical comparison with partial access and as a counterfactual for a further expansion of investability in partially open markets.

In the full-access economy, all securities are available for international trade. In addition,

λ_H foreign individual investors enter the domestic market. Foreign institutional investors may allocate data-processing capacity across both types of risky assets. The entry of foreign individual investors also lowers the average density of private information.

The full-access equilibrium specializes Definition 1 to $\mathcal{L} = \{I, J, K, H\}$, with $\mathcal{Q}_L = \mathbb{R}^2$ for every investor type.

D.1. Full-access equilibrium

The optimal portfolio choices have the same form as in the partial-access model. The optimal data choice problem of domestic institutional investors does not change, but that of foreign institutional investors is no longer trivial because they can now also invest in the type-II risky asset. They therefore allocate $\bar{M}$ data-processing capacity across both types of risky assets, as domestic institutional investors do. The form of the price coefficients changes with the set of investor types allowed to hold each risky asset.

Under full access both assets have the same eligible-investor mass $\lambda_3 \equiv \lambda_I + \lambda_J + \lambda_K + \lambda_H$. For $N \in \{\mathrm{I}, \mathrm{II}\}$, the own-asset SNR and the asset-specific average posterior precision are

$$\zeta_{(N);\mathrm{fu}} = -\frac{r - g_N}{\rho r}\left[(\lambda_I \varphi^2 M_{I,(N);\mathrm{fu}} + \lambda_K M_{K,(N);\mathrm{fu}})\Pi_{e,(N)} + \lambda_1 \Omega_{\epsilon,(N)}\right], \tag{12}$$

$$\bar{\hat{\Omega}}_{y,(N);\mathrm{fu}} = \Omega_{y,(N)} + \hat{\Omega}_{p,(N);\mathrm{fu}} + \frac{\lambda_I \varphi^2 M_{I,(N);\mathrm{fu}} + \lambda_K M_{K,(N);\mathrm{fu}}}{\lambda_3}\Pi_{e,(N)} + \frac{\lambda_1}{\lambda_3}\Omega_{\epsilon,(N)}, \tag{13}$$

where $\hat{\Omega}_{p,(N);\mathrm{fu}} = \zeta^2_{(N);\mathrm{fu}}\Omega_{x,(N)}$. Unlike partial access, no portfolio-constraint Schur discount appears because both risky assets are tradable by foreign institutions.

D.2. Comparative statics of full access

Full access influences the SNRs in a slightly different way. Once the previously inaccessible asset becomes tradable, information produced by the access-restricted group is aggregated directly in that asset's market. In the domestic–foreign implementation, this is information that foreign institutional investors have learned about Type-II risky assets. Domestic institutional investors may change their data acquisition accordingly. The change in average posterior precision from partial to full access contains both a price-information component and a change in the density of privately held information.

Proposition 3 (Market access, price informativeness, and posterior precision). *Market access need not affect price informativeness and average posterior precision in the same direction.*

(i) When Type-II shares become newly investable, their price informativeness can decline if the loss of Type-II information previously transmitted through Type-I prices exceeds the gain from incorporating that information directly into Type-II prices.

(ii) Under partial access, Type-I price informativeness can increase even as the average posterior precision of investors eligible to hold Type-I shares declines.

Thus, price informativeness is an equilibrium aggregation outcome: it need not be monotone in investability or coincide with the information held by investors.

The exact conditions and proofs are collected in Appendix A.

V. Portfolio Constraints and Endogenous Participation

The preceding section isolates how market access changes information allocation and price aggregation when investors can take unconstrained positions. This section introduces a short-sale constraint. With this constraint, access to an asset does not imply that an investor holds it: information affects both the position an investor wishes to take conditional on participation and the probability of participating in that asset market. These participation decisions alter aggregate risk-bearing capacity and, through market clearing, the information content of prices.

With a short-sale constraint, an investor eligible to trade both risky assets need not take a positive position in both. Depending on posterior expected excess payoffs and precision, the investor may hold both assets, only Type I, only Type II, or neither; denote these portfolio states by 11, 10, 01, and 00, respectively. Because the assets are correlated, the decision to hold one asset generally depends on beliefs about the other. The constrained portfolio is therefore not obtained simply by setting negative components of the unconstrained portfolio equal to zero: when one asset is excluded, the remaining position is reoptimized. Information consequently affects constrained demand along two margins: the amount of risk borne conditional on participation and the probability of participating in each asset market.

Under partial access, the access-unrestricted group may take nonnegative positions in both assets, whereas the access-restricted group may take a nonnegative position only in Type I. Its zero position in Type II reflects an investability restriction, not a binding short-sale constraint. In

the domestic–foreign implementation, the foreign investor's effective Type-I precision remains the Schur complement, but information now also affects the probability of taking a positive Type-I position. The short-sale constraint therefore changes both the private value of information and aggregate risk bearing.

Exact constrained aggregation makes the global price map nonlinear. The analysis therefore retains exact portfolio choice and market clearing, while approximating only the price-information map locally around the reference state. The same participation margin applies under closed and full access: closed access isolates domestic participation, partial access adds foreign participation in Type I, and full access makes foreign participation endogenous in both asset markets.

When the short-sale constraint is nonbinding for every eligible investor and asset, the constrained economy coincides with the unrestricted benchmark. The Internet Appendix provides the KKT derivations, the full fixed-point system, regularity conditions, and the full-access specialization.

VI. Welfare Analysis

Having characterized decentralized data acquisition, this section compares it with the allocation chosen by a planner and identifies the wedge between private and social marginal values. The comparison is conditional on a fixed access and trading regime: the planner preserves market access and the short-sale constraints, so the section evaluates the constrained efficiency of the information allocation within a given regime, not the welfare effect of changing market access itself. Within that fixed regime the welfare question is general: access restrictions can make privately valuable information about one asset affect price formation and allocation in another.

The cross-asset learning mechanism does not by itself establish a welfare loss. To evaluate efficiency, consider a constrained second-best planner that chooses only the common data allocation of each institutional-investor type, taking market access, short-sale constraints, private signals, decentralized portfolio choice, and rational-expectations market clearing as given. The Internet Appendix provides the complete welfare derivation and formulation.

Let a denote the access tag, with cl for closed access, no access tag for the partially open benchmark, and fu for full access. Because this section preserves the short-sale constraint, generic decentralized welfare objects carry the additional tag c and planner allocations append pl. Thus

$\mathrm{CE}_{L;a,\mathrm{c}}(\boldsymbol{m})$ is type L's ex ante certainty equivalent in the constrained decentralized equilibrium under access regime a, and the corresponding planner allocation carries $a, \mathrm{c}, \mathrm{pl}$. The vector $\boldsymbol{m}$ collects institutional data allocations, and $\mathcal{L}_a$ denotes the set of active investor types in regime a—the set $\mathcal{L}^R$ of Section III, re-indexed by the access tag a in place of the regime label R.

We consider one planner objective at a time. The global planner weights every active investor type and solves

$$\boldsymbol{m}_{G;a,\mathrm{c},\mathrm{pl}} \in \arg\max_{\boldsymbol{m}} \sum_{L \in \mathcal{L}_a} \lambda_L \, \mathrm{CE}_{L;a,\mathrm{c}}(\boldsymbol{m}), \tag{14}$$

whereas the domestic regulator values only domestic investors and solves

$$\boldsymbol{m}_{D;a,\mathrm{c},\mathrm{pl}} \in \arg\max_{\boldsymbol{m}} \left[\lambda_I \, \mathrm{CE}_{I;a,\mathrm{c}}(\boldsymbol{m}) + \lambda_J \, \mathrm{CE}_{J;a,\mathrm{c}}(\boldsymbol{m})\right]. \tag{15}$$

Both are subject to the same capacity, portfolio, price-consistency, and market-clearing conditions as the decentralized economy. In the main benchmark, information-processing capacity is an exogenously committed budget: for every data-producing type L, $M_{L,(\mathrm{I})} + M_{L,(\mathrm{II})} = \bar{M}$. Thus the planner can reallocate capacity across the two assets but cannot leave it idle. This restriction preserves the paper's allocation problem and isolates whether decentralized investors direct a given processing budget to the socially efficient source.

An atomistic investor takes the price law as given when choosing data. For either planner objective, the planner additionally internalizes how a common change in type L's data allocation changes prices, posterior moments, and the probability that trading constraints bind. Denote this omitted marginal equilibrium effect by $\mathcal{X}_{L;a,\mathrm{c}}$. At an interior decentralized allocation, the private marginal value is zero, so

$$\left.\frac{dW_{a,\mathrm{c}}}{dm_L}\right|_{\boldsymbol{m}_{a,\mathrm{c}}} = \mathcal{X}_{L;a,\mathrm{c}}(\boldsymbol{m}_{a,\mathrm{c}}). \tag{16}$$

Consequently, a nonzero $\mathcal{X}_{L;a,\mathrm{c}}$ establishes that the decentralized information allocation is not constrained efficient.

For the partial-opening mechanism, let $n_K = M_{K,(\mathrm{II})}$ be the foreign information-capacity choice assigned to the inaccessible asset. Its decentralized value under partial access is $n_{K;\mathrm{c}}$. By the wedge condition of (16), since $\mathcal{X}_{K;\mathrm{c}}(\boldsymbol{m}_{\mathrm{c}}) < 0$—equivalently $-\mathcal{X}_{K;\mathrm{c}}(\boldsymbol{m}_{\mathrm{c}}) > 0$—a marginal reduction in n_K raises welfare; under strict concavity, the global planner chooses less foreign processing of the inaccessible asset. The following proposition states this partial-access cross-

asset result in a specialized fixed-active-set benchmark and reports the wedge in closed form. In that benchmark, c denotes the posterior cross-precision of the foreign investor's signal about the two assets, $\Pi_{e,(n)} = \beta^2 \Omega_{e,(n)}$ the effective data precisions, and $0 \leq \chi \leq 1$ the endogenous social pass-through of the private cross-learning benefit, i.e. the fraction of that benefit that remains in weighted social surplus.

Proposition 4 (Global planner and the non-investable asset). *In the fixed-active-set benchmark, when both the decentralized and the global-planner data choices are interior and $c \neq 0$, the global planner allocates weakly less capacity to information about the non-investable asset, and*

$$n_{K;\mathrm{c}} - n_{K;G,\mathrm{c,pl}} = \frac{|c| \left(1 - \sqrt{\chi}\right)}{\sqrt{\Pi_{e,(\mathrm{I})} \Pi_{e,(\mathrm{II})}}}. \tag{17}$$

The wedge is positive when fundamentals are correlated and $\chi < 1$, vanishes when correlation is zero, and shrinks when broader access raises the social pass-through χ. The Internet Appendix derives this result. Equations (16)–(17) therefore provide conditional analytical predictions rather than a calibrated sign restriction. The quantitative section evaluates the decentralized and planner allocations under the estimated primitives and reports the associated certainty-equivalent welfare gains separately.

VII. Quantitative Analysis

This section takes the model to the Shanghai–Hong Kong Stock Connect setting. It first maps the institutional environment into the calibration, then evaluates the benchmark, policy counterfactuals, and reduced-form evidence. More broadly, the framework applies to settings in which investors differ in market access, data acquisition, and trading constraints.

A. Institutional setting

The domestic Chinese stock market (the "A-share" market) started in 1990 with the establishment of two independently operated domestic stock exchanges: the Shanghai Stock Exchange (SSE) and the Shenzhen Stock Exchange (SZSE). Historically, the Chinese stock market has been only weakly integrated with international markets. The Shanghai–Hong Kong Stock Connect Program was an important step in China's capital-market liberalization. Launched on November 17, 2014,

it allows local and international investors in Hong Kong to trade SSE shares (northbound trading) through the Hong Kong Stock Exchange (HKEX). Shanghai–Hong Kong Stock Connect primarily covers large-capitalization index constituents listed on the SSE and HKEX. The Internet Appendix provides the trading arrangement, eligibility criteria, and international comparisons.

China's gradual liberalization provides a useful setting because prior evidence suggests that Chinese stock prices have become more informative about future profits over time (Carpenter et al., 2021). This paper studies how changes in market access can reshape information allocation and price aggregation within that broader institutional evolution.

B. Parameterization and Estimation

The calibration proceeds in three steps. First, the investor-type shares and the risk-free rate are calculated directly from the data. Second, the price coefficients are estimated by OLS from cross-sectional regressions. The estimated matrix $\widehat{\boldsymbol{C}}$ provides the price-loading moment residuals, while $\widehat{\boldsymbol{B}}$ is used entirely in a preliminary step to recover the dividend-growth parameters $\boldsymbol{G}$ through $\boldsymbol{G} = r\widehat{\boldsymbol{B}}(\boldsymbol{I} + \widehat{\boldsymbol{B}})^{-1}$; with $\boldsymbol{G}$ fixed, the model-implied $\boldsymbol{B}$ coincides with $\widehat{\boldsymbol{B}}$ by construction, so $\widehat{\boldsymbol{B}}$ residuals carry no identifying information and are excluded from the third-stage objective. Third, the remaining structural parameters are estimated by weighted minimum distance, minimizing the price-loading ($\widehat{\boldsymbol{C}}$), price-noise-covariance, ownership-share, and holdings-ratio residuals, all evaluated at the fully constrained local equilibrium with the data choices solved endogenously as an inner fixed point and the short-sale constraints imposed asset by asset.

Identification The different sets of moments identify distinct components of the model. Ownership-share moments identify the relative investor masses $\boldsymbol{\lambda}$ after accounting for equilibrium portfolio demand. Price-loading moments identify the covariance structure of information and noise, because endogenous data acquisition affects equilibrium price sensitivities. Price-noise moments discipline the supply-noise parameters conditional on the equilibrium price signal. The foreign cross-asset precision object z is an endogenous equilibrium outcome rather than an additional data moment; the spillover channel is disciplined through the price coefficients and the equilibrium structure.

B.1. Measured inputs and calibration targets

The observed investor-type shares and the risk-free rate are measured inputs rather than estimated parameters: the shares are calibration targets for the investor masses $\boldsymbol{\lambda}$, and r is a fixed input. The scale of the information technology is set by three further choices: $\beta_I = 1$ and $\bar{M} = 2$ are normalizations, whereas $\beta = 3$ (equivalently $\varphi = \beta_I/\beta = 1/3$) is a substantive fixed technology assumption. The supplementary analyst-forecast exercise is informative against equal processing efficiency but does not sharply identify the magnitude of the gap: joint or conditional re-estimation improves fit only by moving risk aversion to an economically extreme region. We therefore retain $\beta = 3$ as the maintained benchmark and report the forecast construction, profiles, and joint estimates in the Internet Appendix.

Market share The aggregate data on the composition of shareholders are obtained from China Securities Regulatory Commission (CSRC). The largest holder of Chinese A-shares is industrial capital, including non-financial companies and general legal persons. Considering that industry capital and original shareholders' holdings cannot be freely circulated[3], they are not included in the analysis. In the same vein, market shares are calculated with *negotiable market capitalization*, i.e. market capitalization for tradable common listed shares only.

In 2013, individual investors held more than 33% of total A-share market capitalization, peaking above 39% in 2015 and declining below 30% by 2023; domestic institutional investors held around 15%, and foreign capital about 3%. These ratios are computed against total market capitalization, which includes non-tradable shares predominantly held by industrial capital and original shareholders. Since only negotiable market capitalization is used in this model, these shares are restated on the negotiable basis: for the time window surrounding the launch of the Shanghai-Hong Kong Stock Connect, domestic individual investors, domestic institutional investors, and foreign institutional investors hold 66%, 28%, and 6% of negotiable market capitalization, respectively, which sum to one.

Risk-free rate The risk-free rate is set to the gross return $r = 1.042$, corresponding to a mean yield to maturity of approximately 4.2% per annum on the 10-year government bond in 2014.

[3]These are common shares held by state legal persons or original promoters. They cannot be traded on the exchange but can be traded by auction or transfer by agreement under the approval of the CSRC.

The government bond yield curve is published by China Central Depository & Clearing Co., Ltd. (CCDC).

B.2. OLS estimation for price coefficients

Data on daily stock prices and the eligibility details of the Stock Connect Programs are obtained from the China Stock Market & Accounting Research (CSMAR) database. Following the measurement methodology in the literature, unexpected earnings proxy for the dividend innovation $\widetilde{y}$. Earnings surprises are proxied by residuals from the corresponding time-series forecasting models. To estimate the price coefficients, we run the following cross-sectional regressions:

$$ma_{i,t,(\mathrm{I})} = \hat{A}_{t,(\mathrm{I})} + \hat{B}_{t,(\mathrm{I})} ea_{i,t-1,(\mathrm{I})} + \hat{C}_{t,(\mathrm{I})} ue_{i,t,(\mathrm{I})} + \hat{C}^{(c)}_{t} ue_{i,t,(\mathrm{II})} + \hat{K}_{i,t,(\mathrm{I})} Z_{i,t,(\mathrm{I})} + \hat{\varepsilon}_{i,t,(\mathrm{I})},$$
$$ma_{i,t,(\mathrm{II})} = \hat{A}_{t,(\mathrm{II})} + \hat{B}_{t,(\mathrm{II})} ea_{i,t-1,(\mathrm{II})} + \hat{C}_{t,(\mathrm{II})} ue_{i,t,(\mathrm{II})} + \hat{K}_{i,t,(\mathrm{II})} Z_{i,t,(\mathrm{II})} + \hat{\varepsilon}_{i,t,(\mathrm{II})},$$

where $ma_{i,t}$ is the logarithm of market capitalization divided by total assets; $ea_{i,t-1}$ is the previous year's EPS times shares outstanding, divided by total assets; $ue_{i,t}$ is the earnings surprise normalized by total assets; and $Z_{i,t}$ is a collection of dummies at the 2012 CSRC sub-industry level. The Internet Appendix provides detailed variable construction. Because the outcome is the log of the market-to-assets ratio, the estimated coefficients are semi-elasticities of market capitalization with respect to earnings and earnings surprises—the units in which the equilibrium price-loading moments are matched. These regression coefficients provide empirical moments corresponding to the model's equilibrium price sensitivities $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{C}$, rather than direct estimates of structural coefficients. The residuals are used as an empirical proxy for the price-noise component $\boldsymbol{D}\boldsymbol{\Sigma}_x\boldsymbol{D}'$, in the sense that any omitted fundamental, measurement, or misspecification component orthogonal to the controls is also absorbed into the residual covariance.

The supply-noise precisions $\Omega_{x,(\mathrm{I})}$ and $\Omega_{x,(\mathrm{II})}$ are therefore jointly identified with the other structural parameters under this measurement assumption rather than observed directly. The Internet Appendix reports a robustness exercise that re-weights the price-noise block and leaves the estimates and the allocation corners unchanged; the exercise demonstrates robustness to the block's weight rather than validation of this maintained measurement mapping, because the regression residual covariance also absorbs omitted fundamentals, measurement error, and misspecification.

The growth matrix $\boldsymbol{G}$ is identified directly from the estimated equilibrium price-loading ma-

trix $\widehat{\boldsymbol{B}}$ through the restriction $\boldsymbol{B} = (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}$. This identification follows from the present-value restriction linking equilibrium prices to expected dividend growth rather than from an independent estimate of future growth rates. The estimates yield $g_1 > g_2$. This ordering is economically plausible given the selection of foreign-investable securities toward large, liquid, and relatively strong-performing firms, although it is not imposed by the model and should not be interpreted as a mechanical consequence of market access itself.

B.3. Weighted minimum-distance estimation

Let $m_L(\boldsymbol{\lambda})$ denote the model-implied market share of type-L investors,

$$m_L(\boldsymbol{\lambda}) \equiv \frac{\lambda_L \, \boldsymbol{q}_L' \boldsymbol{p}}{\sum_{L'} \lambda_{L'} \boldsymbol{q}_{L'}' \boldsymbol{p}}, \qquad L \in \{I, J, K\}. \tag{18}$$

Both price weights and market shares are homogeneous of degree zero in $\boldsymbol{\lambda}$, so it is convenient to normalize $\lambda_I + \lambda_J + \lambda_K = 1$.

Treating the observed market shares of each investor type as the exogenous weights λ_L is problematic. This mapping is exact only when equilibrium demands are identical across types; with information-driven dispersion in demands, the observed market shares are equilibrium objects that also embed the cross-type dispersion of posterior expectations. Therefore $\boldsymbol{\lambda} = (\lambda_I, \lambda_J, 1 - \lambda_I - \lambda_J)$ is to be estimated by moment matching.

Because price informativeness and posterior variances depend on endogenous data choices, each candidate parameter vector is evaluated at the fixed point of the constrained partial-access equilibrium. The calibration matches price loadings, price-noise covariance, three investor ownership shares, and the domestic institutional holdings ratio. Effective data precision $\boldsymbol{\Pi}_e$ is estimated directly, while $\beta = 3$, $\beta_I = 1$, and $\bar{M} = 2$ are maintained benchmark choices.

The Type-II effective-precision parameter is set-identified at the estimated corner because no investor processes Type-II data there; we report a corner-supporting evaluation point and do not interpret its level as a point-identified economic fact. The holdings-ratio moment disciplines the domestic allocation through equilibrium restrictions. The full moment vector, weighting, local-rank diagnostics, and joint-estimation robustness exercises for relative processing efficiency are in the Internet Appendix. Table I reports the estimated primitives and fit. Ownership shares are equilibrium outcomes rather than investor-mass parameters, and institutional information choices are solved endogenously within estimation.

Calibration results The parameter estimates, targeted moments, endogenous information allocation, and participation outcomes are summarized in Tables I and II. As an untargeted diagnostic, the estimated model reproduces the direction of the data-based price variance reduction: 9.32% versus 12.37% in the data for type-I shares and 4.15% versus 4.40% for type-II shares. Because this moment is not separately targeted in estimation, the approximate agreement is reassuring; it is not, however, an independent out-of-sample validation, since the diagnostic is built from the same price-information objects that underlie the targeted $\boldsymbol{C}$ and price-noise moments. The residual tension in Table I now concentrates in the price-noise moments: the model's $\Sigma_{11}^{p,\text{noise}}$ is 0.3008 against data of 0.23479 (marginal residual 3.00) and $\Sigma_{22}^{p,\text{noise}}$ is 0.2257 against 0.32837 (marginal residual -3.51), while the own-price loadings fit well, with C_{22} at 2.2874 against 2.2683 (marginal residual 0.07). The reparameterized estimation moves the residual tension from the non-investable own-price loading toward the price-noise covariance; a fuller account of this trade-off appears in the Internet Appendix sensitivity discussion. Re-weighting the price-noise block at half its sampling weight and re-estimating leaves every parameter estimate and the allocation corners unchanged (Internet Appendix), so the imperfect fit of these two moments does not drive the estimated information allocation. A related point concerns the cross-loading: the reduced-form cross-price coefficient is positive ($C_{12}^{\text{data}} = 1.1766$), yet the estimated structural equilibrium generates a much smaller cross-loading ($C_{12}^{\text{model}} = 0.0632$). This is not a statistical failure (marginal residual -1.05), and it is economically coherent: investors internalize correlated fundamentals through private data acquisition, so the incremental informational content of prices for the cross fundamental is small even though the reduced-form comovement—which also reflects discounting and equilibrium trading—is sizable.

C. Benchmark Equilibrium and Information Mechanism

In the calibrated model, foreign institutional investors hold a 6.02% share of negotiable market value (Table I), close to the 6% observed in the data. The first quantitative question is why granting foreign investors access only to Type-I shares can nevertheless affect market outcomes for both Type-I and Type-II shares, and why the two responses need not be alike. Table II collects the equilibrium objects behind this transmission. Foreign entry directly changes informed demand and risk bearing in Type I. Because the resulting Type-I price is a public signal about correlated fundamentals, domestic investors update beliefs, reallocate data-processing capacity, and adjust

Table I
Structural parameter estimates and benchmark model fit.

Panel A. Structural parameters

Panel A reports the economically central primitives of the estimated partial-access economy with short-sale constraints. $\Pi_{e,(\mathrm{II})}$ is a corner-supporting, set-identified value; the Internet Appendix characterizes its identified region and reports the maintained technology choices, remaining parameters, and full identification diagnostics.

Parameter	Economic interpretation	Value
$\Pi_{e,(\mathrm{I})}$	Effective data precision, Type I	7,438.0460
$\Pi_{e,(\mathrm{II})}$	Effective data precision, Type II (corner-supporting)	73.8648
$\Omega_{\epsilon,(\mathrm{I})}$	Domestic soft-information precision, Type I	1,596.7535
$\Omega_{\epsilon,(\mathrm{II})}$	Domestic soft-information precision, Type II	1,108.4729
ρ	Risk aversion	1.4613
λ_K	Foreign institutional investor mass	0.0195

Panel B. Targeted moments and benchmark fit

Panel B reports the headline targeted moments. The Internet Appendix reports the full moment vector, weighting, sampling standard errors, marginal residuals, objective, and robustness diagnostics.

Moment	Data	Model
C_{11}	3.7123	3.6565
C_{12}	1.1766	0.0632
C_{22}	2.2683	2.2874
$\Sigma_{11}^{p,\text{noise}}$	0.23479	0.3008
$\Sigma_{12}^{p,\text{noise}}$	0.00007	0.0040
$\Sigma_{22}^{p,\text{noise}}$	0.32837	0.2257
Foreign institutional ownership share	0.0600	0.0602
Domestic institutional Type-I/II holdings ratio	1.5953	1.6232

Table II
Benchmark equilibrium and information mechanism.

Values refer to the estimated partial-access equilibrium with short-sale constraints. Panels report ownership and participation, information technology and capacity allocation, zero-holding probabilities, and variance reduction relative to the prior. Capacity is measured relative to $\bar{M} = 2$; the active foreign mass equals λ_K times the Type-I participation probability.

	Benchmark value
Panel A. Ownership, mass, and participation	
Foreign institutional ownership share	6.02%
Estimated foreign institutional model mass, λ_K	1.95%
Probability of a zero type-I holding	92.8%
Type-I participation probability	7.2%
Participating foreign mass, $\lambda_K \times 7.2\%$	0.14%
Panel B. Information technology and allocation	
Foreign processing technology, β	3.000
Domestic institutional processing technology, β_I	1.000
Relative domestic processing efficiency, $\varphi \equiv \beta_I/\beta$	1/3
Effective precision per processed observation, foreign relative to domestic	9×
Domestic capacity allocated to Type I / Type II	2.000 / 0.000
Foreign capacity allocated to Type I / Type II	2.000 / 0.000
Panel C. Probability of a zero holding	
Domestic institutions, I: Type I / Type II	77.7% / 73.7%
Domestic individuals, J: Type I / Type II	72.8% / 73.7%
Foreign institutions, K: Type I	92.8%
Panel D. Market outcomes	
Price-based variance reduction, Type I / Type II	9.32% / 4.15%
Total variance reduction, Type I / Type II	83.97% / 67.23%

portfolios across both asset types. Partial access can therefore generate market-wide and heterogeneous effects even though foreigners directly trade only one segment. The small foreign sector reinforces this mechanism: its estimated mass is $\lambda_K = 0.0195$ (1.95% of the investor population), and its endogenous Type-I participation probability is just 7.2%, so the actively participating foreign mass is approximately 0.14% of the population. Ownership, investor mass, active participation, and informational influence are distinct equilibrium objects.

The first reason is that observed ownership is an equilibrium outcome, not the foreign investor mass. Market shares are determined by equilibrium holdings (18): a low observed share can reflect a high probability of a zero holding induced by portfolio constraints, smaller positive positions conditional on participation, differences in beliefs and posterior precision, or asset-access restrictions. The calibration estimates a foreign institutional model mass of $\lambda_K = 0.0195$, or 1.95% of the investor population (Table II), even though foreign institutions hold only 6.02% of market value in the benchmark equilibrium. The gap reflects the short-sale constraint, which binds frequently: the type-I zero-holding probability is 92.8% (Table II), so the participation probability is only 7.2%, and the participating foreign mass is approximately $0.0195 \times 0.072 = 0.14\%$. The estimated foreign mass is itself small, and infrequent participation makes the actively trading foreign sector smaller still. Ownership share, investor mass, and informational influence are three distinct objects: even a quantitatively tiny participating foreign mass can exert disproportionate informational influence if its information is sufficiently productive and is transmitted through equilibrium prices. This conclusion is conditional on the benchmark identification region: an almost equally fitting specification with equal type-I and type-II effective precision (objective 1.2919 versus 1.2587) flips the domestic information allocation to type-II and eliminates the type-I foreign-entry effect (Internet Appendix), so the data do not yet globally select this mechanism.

The second reason is that foreign institutions have a processing advantage under the maintained benchmark technology: $\beta = 3$, whereas domestic institutional technology is normalized to $\beta_I = 1$ (Table II), so $\varphi = 1/3$ and $\varphi^2 = 1/9$. The supplementary forecast-error exercise does not sharply identify this gap: appending the moment can improve fit only by moving risk aversion to an economically extreme region, and the profile is nearly flat over a wide range of processing advantages. We therefore retain $\beta = 3$ as a maintained benchmark and report the forecast construction, profile, and joint-estimation results in the Internet Appendix. With effective data precision $\mathbf{\Pi}_e \equiv \beta^2 \mathbf{\Omega}_e$, one correctly processed observation contributes $\mathbf{\Pi}_e$ to posterior

precision when processed by a foreign institution and $\varphi^2\mathbf{\Pi}_e$ when processed by a domestic institution: holding the data source fixed, a foreign-processed observation delivers nine times the effective precision of the same observation processed with the domestic institutional technology. A marginal foreign investor is therefore informationally different from a marginal domestic holder. In the benchmark, both foreign and domestic institutions allocate their entire processing capacity to type-I data (Table II); this concentration is consistent with the targeted ratio of domestic institutional holdings in type-I to type-II shares of about 1.60 in the data and 1.62 in the model. A small ownership stake can accordingly be associated with a concentrated, high-precision contribution to the information environment of the investable market.

Third, foreign information does not remain private to the foreign sector. Foreign informed demand transmits part of that information into equilibrium prices, and the price is a public signal observed by all investors: foreign entry and foreign data processing shift informed demand, which changes the equilibrium price signal, which updates domestic posterior beliefs, which alter domestic information allocation and portfolio demand, and thereby feed back into the price. A foreign sector can therefore have market-wide effects even when its direct holdings are small, because it affects the marginal price signal rather than the average ownership share. This is fundamentally different from a partial-equilibrium ownership-share calculation.

Fourth, domestic institutions do not hold their data acquisition fixed when foreign investors enter. Their processing capacity is allocated endogenously across the two asset types, and a change in price informativeness changes the marginal value of private data. Foreign entry therefore has both a direct information effect, through foreign processing and informed demand, and an indirect crowding-in or crowding-out effect through the reoptimization of domestic data acquisition. The equilibrium effect of foreign entry need not be proportional to the foreign ownership share.

The high zero-holding probabilities reported in Table II (Panel C) warrant emphasis, because they are central to the large short-sale effects documented below. The short-sale constraint binds frequently in the estimated economy: Table II (Panel C) reports zero-holding probabilities of 77.7% and 73.7% for domestic institutions, 72.8% and 73.7% for domestic individuals, and 92.8% for foreign institutions in type-I; equivalently, the positive-holding probabilities are 22.3%, 26.3%, 27.2%, 26.3%, and 7.2%, respectively. The combination that looks most extreme—foreign institutions allocate their entire processing capacity to type-I data ($M_{K,(\mathrm{I})} = 2$) and yet hold type-I shares only about 7% of the time—is economically coherent rather than contradictory. Informa-

tion has option value: it raises posterior precision in the states in which the constraint is slack and the investor participates, so a low participation probability need not imply a low ex ante value of data. More generally, the information choice is evaluated ex ante over all states, whereas portfolio participation is determined ex post after observing signals and prices; the two decisions therefore need not move together. The high binding probabilities are a property of the estimated primitives (large position-size exposures relative to the aggregate supply noise), not an input. Because the estimated binding probabilities are high, the Internet Appendix reports sensitivity analyses assessing the quantitative importance of the short-sale constraint.

Fifth, correlated fundamentals and domestic general-equilibrium responses transmit the effect across asset types. Although foreign institutions process only Type-I data in the calibrated benchmark, changes in foreign Type-I informed demand alter the price-information environment faced by domestic investors, who learn about and trade both assets. The model therefore explains why the effects of foreign access need not be confined to the shares that foreigners are permitted to hold.

Finally, foreign entry also changes the composition of the information-bearing investor population in the opposite direction. Foreign institutions have superior hard-information processing but no domestic soft information, so their entry simultaneously adds precise processed hard information and dilutes the average density of domestic soft information. The net effect on market outcomes is therefore not determined by the foreign ownership share alone; it depends on the interaction of the foreign mass, processing technology, soft-information precision, data allocation, asset access, and portfolio constraints.

These mechanisms clarify why the effect of partial access need not be confined to the securities that foreigners can hold and why the two asset types can respond differently. In the constrained closed-to-partial comparison, price-based variance reduction changes from 8.85% to 9.32% for Type I (about a 5.3% increase relative to the closed-market level) and from 4.18% to 4.15% for Type II (Table III). The price-information effect is therefore positive but modest for Type I and essentially flat for Type II in this particular statistic. The broader mechanism is nevertheless market-wide: foreign entry changes the Type-I price signal, and that public signal feeds into domestic beliefs, data allocation, and portfolios in both markets. The 0.14% actively participating foreign mass illustrates why direct participation is not a sufficient statistic for that transmission. The much larger gains documented in the unrestricted counterfactual below arise primarily from relaxing the short-sale constraint rather than from incremental foreign investability.

D. Counterfactual Market Opening

The benchmark shows how partial access can transmit foreign entry beyond the directly investable segment through equilibrium prices and domestic responses. We next use the estimated model to ask what would happen if foreign institutions could trade both share types. Holding the estimated Chinese primitives fixed, we compare two scenarios: full access with a short-sale constraint maintained, in which foreign institutions can trade both Type-I and Type-II shares but all risky-asset positions remain nonnegative; and full access without a short-sale constraint. Table III reports the outcomes, together with the partial-access benchmark for comparison.

The comparison shows that full access has sharply different implications depending on the portfolio regime. When short sale is allowed, the full-access equilibrium is well behaved: foreign institutions allocate all of their processing capacity to Type-I information, and prices remove 65.92% and 37.99% of prior variance about Type-I and Type-II fundamentals, respectively (Table III, Panel D). With a short-sale constraint, the constrained full-access equilibrium converges to a pure symmetric data-choice fixed point at the current calibration (Internet Appendix Section IA.H); foreign processing remains concentrated on Type-I data ($M_{K,(\mathrm{I})} = 2.000$ and $M_{K,(\mathrm{II})} = 0.000$), but the constraint severely impedes the aggregation of this private information into prices: price-based variance reduction falls to 9.32% for Type I and 4.15% for Type II, and total variance reduction is 83.97% and 66.81%.

The short-sale constraint therefore does more than restrict foreign portfolios; it changes the private value of information. The calibration identifies the short-sale margin as the dominant driver of the large full-access numbers: expanding investability from partial to full access leaves price informativeness essentially unchanged within either trading regime (constrained full access delivers the same $9.32\%/4.15\%$ price-based variance reductions as the partial-access benchmark, and the unrestricted partial- and full-access equilibria are likewise indistinguishable), whereas removing the short-sale constraint raises Type-I price-based variance reduction from 9.32% to 65.92%. The calibration also draws a sharp distinction between information possessed by investors and information revealed through equilibrium prices: a market can contain substantial private information while prices reveal only a small fraction of it when portfolio constraints weaken informed risk bearing. Portfolio constraints primarily impede the aggregation of private information into prices rather than altering the concentrated Type-I information allocation itself. The effect of relaxing the short-sale constraint is much larger than the effect of incremental

Table III
Market-access and short-sale counterfactuals.

All counterfactuals hold estimated Chinese primitives fixed. Closed removes foreign investors; constrained full access permits foreign Type-II trading but retains a short-sale constraint; Panel D also permits short sale. Panel E reports constrained-minus-unrestricted differences in percentage points. Both full-access equilibria converge; entries below 0.0005 in absolute value are reported as zero.

	Closed Counterfactual	**Partial (Estimated)**	Full access† (Constrained)
Panel A. Information production			
Domestic $M_{I,(\mathrm{I})}$	2.000	2.000	2.000
Domestic $M_{I,(\mathrm{II})}$	0.000	0.000	0.000
Foreign $M_{K,(\mathrm{I})}$	n.a.	2.000	2.000
Foreign $M_{K,(\mathrm{II})}$	n.a.	0.000	0.000
Panel B. Price-based variance reduction			
Type I, VR^{p}_{I}	8.85%	9.32%	9.32%
Type II, VR^{p}_{II}	4.18%	4.15%	4.15%
Panel C. Total variance reduction			
Type I, $VR^{\mathrm{tot}}_{\mathrm{I}}$	82.32%	83.97%	83.97%
Type II, $VR^{\mathrm{tot}}_{\mathrm{II}}$	67.23%	67.23%	66.81%
Panel D. Full access, short sale allowed			
Foreign capacity allocated to Type I			100.0%
Foreign capacity allocated to Type II			0.0%
Type I, VR^{p}_{I}			65.92%
Type II, VR^{p}_{II}			37.99%
Type I, $VR^{\mathrm{tot}}_{\mathrm{I}}$			87.60%
Type II, $VR^{\mathrm{tot}}_{\mathrm{II}}$			72.04%
Panel E. Effect of a short-sale constraint (constrained minus unrestricted, pp)			
$\Delta VR^{p}_{\mathrm{I}}$	−51.56	−56.60	−56.60
$\Delta VR^{p}_{\mathrm{II}}$	−34.71	−33.84	−33.84
$\Delta VR^{\mathrm{tot}}_{\mathrm{I}}$	−3.56	−3.63	−3.63
$\Delta VR^{\mathrm{tot}}_{\mathrm{II}}$	−5.29	−5.11	−5.23

†Full access with a short-sale constraint maintained.

investability in either access regime. The quantitative weight therefore falls on the portfolio-constraint margin at the Chinese benchmark. We next evaluate the efficiency of the resulting decentralized information allocation.

E. Information Allocation Efficiency

For compactness, the tables report variance reduction relative to the prior. Price-based variance reduction uses the posterior based on the price signal alone, while total variance reduction uses a population-wide posterior-precision statistic with a common population denominator. This quantitative statistic is distinct from the asset-eligible average posterior precision $\bar{\hat{\Omega}}_{y,(N)}$ used in the analytical comparative statics above. For the population-wide statistic, full symmetric individual posterior precision matrices are averaged before inversion; both variance-reduction statistics invert a full precision matrix before taking its diagonal because dividend innovations are correlated.

F. Reduced-Form Evidence

The structural analysis characterizes how market access affects equilibrium information allocation and prices. This subsection documents whether the Shanghai–Hong Kong Stock Connect announcement and launch were accompanied by differential changes in information-related market outcomes between Type-I shares, which became foreign-investable, and Type-II shares, which remained inaccessible. These patterns provide an empirical counterpart to the model's focus on how a targeted change in investability can reshape the information environment.

Since market participants could anticipate the policy change before trading started, the empirical design distinguishes the announcement-to-launch interval from the post-launch period and estimates

$$Y_{it} = \alpha_i + \gamma_{j(i),t} + \beta_A(\text{Eligible}_i \times \text{Announcement}_t) + \beta_L(\text{Eligible}_i \times \text{Post}_t) + \varepsilon_{it}, \tag{19}$$

where α_i denotes a firm fixed effect and $\gamma_{j(i),t}$ an industry-by-day fixed effect. The eligible group is fixed using the April 10, 2014 reference universe and retained throughout the sample. The preferred comparison restricts both groups to Shanghai-listed stocks: 562 eligible stocks and 443 never-eligible stocks. This same-exchange comparison avoids confounding the treatment con-

Table IV
Decentralized and efficient information allocation.

The planner chooses a common institutional allocation of capacity $\bar{M}$; prices and private portfolios are re-solved under each row's trading rules. Welfare is the investor-mass-weighted sum of ex ante certainty equivalents. Solutions are checked on a global grid, allocation boundaries, and local refinement. Full $+c$ is the converged constrained full-access equilibrium.

Panel A. Information-allocation wedges

Regime	$\Delta M_{I,\mathrm{I}}$	$\Delta M_{I,\mathrm{II}}$	$\Delta M_{K,\mathrm{I}}$	$\Delta M_{K,\mathrm{II}}$
Closed	-2.000	$+2.000$	n.a.	n.a.
Closed $+c$	-2.000	$+2.000$	n.a.	n.a.
Partial	-2.000	$+2.000$	-1.944	$+1.944$
Partial $+c$	-2.000	$+2.000$	-1.872	$+1.872$
Full	-2.000	$+2.000$	-1.950	$+1.950$
Full $+c$	-2.000	$+2.000$	-1.872	$+1.872$

Panel B. Welfare

Regime	Decentralized W_r	Planner $W_{r,\mathrm{pl}}$	Wedge ΔW_r
Closed	0.8382	0.8898	0.0516
Closed $+c$	0.6502	0.6562	0.0060
Partial	0.8011	0.8923	0.0912
Partial $+c$	0.6440	0.6514	0.0074
Full	0.8018	0.8931	0.0913
Full $+c$	0.6490	0.6561	0.0071

trast with Shanghai–Shenzhen movements. The preferred specification omits contemporaneous firm controls, which can respond to liberalization and are mechanically related to return-based outcomes.

The outcome variable is idiosyncratic volatility, the annualized 30-trading-day standard deviation of market-model residuals. β_A is the announcement-to-launch differential relative to the pre-announcement baseline, β_L is the post-launch differential, and $\beta_L - \beta_A$ is the incremental post-launch contrast. These return-based measures are not direct empirical counterparts of the model's precision objects; the exercise documents differential market outcomes around the opening rather than tests the structural mechanism.

Table V reports the preferred same-exchange specification and a size-adjusted variant. In the preferred specification, eligible stocks have lower idiosyncratic volatility both between announcement and launch (-0.0388, $s.e. = 0.0066$) and after launch (-0.0385, $s.e. = 0.0065$), while the incremental post-launch contrast is indistinguishable from zero. Size adjustment attenuates the announcement-period estimate but leaves the post-launch differential negative. Thus the broad differential is robust, whereas its timing is not cleanly attributable to anticipation versus implementation.

VIII. Conclusion

Market access changes not only who can hold assets but also how investors allocate data-processing capacity and how private information is aggregated into equilibrium prices. When investors differ in soft information and data-processing technology, access restrictions also change the return to acquiring data and the information environment faced by other investors through equilibrium prices.

The theory delivers three main implications. First, market access and information relevance need not coincide: under partial access, foreign institutions may acquire data about an asset they cannot hold because correlated fundamentals make that information useful for trading an accessible asset. Second, this cross-asset learning is an intermediate-precision phenomenon: sufficiently noisy data are not worth acquiring, while sufficiently precise information is increasingly revealed through domestic learning and prices. Third, market access, private information, and price informativeness need not move together. Making an asset newly investable can reduce its price informativeness when the loss of cross-price information exceeds the gain in own-price

Table V
Stock Connect and idiosyncratic volatility.

Estimates compare announcement-eligible with never-eligible Shanghai stocks using equation (19). Column (1) includes firm and industry-by-day fixed effects; column (2) additionally includes pre-treatment size-decile-by-day effects on the common 2013-size sample. Standard errors are two-way clustered by firm and date. The final row is $\beta_L - \beta_A$.

	(1)	(2)
Eligible × Announcement-to-launch period	-0.0388^{***}	-0.0148^{*}
	(0.0066)	(0.0080)
Eligible × Post-launch period	-0.0385^{***}	-0.0277^{***}
	(0.0065)	(0.0078)
Post-launch − Announcement	0.0002	-0.0129^{*}
	(0.0061)	(0.0073)
Sample	Shanghai	Shanghai
Firm fixed effects	Yes	Yes
Industry × day fixed effects	Yes	Yes
Pre-treatment size decile × day fixed effects	No	Yes
Controls	No	No
Clustering	Firm & date	Firm & date
Observations	761,168	723,108
Within R^2	0.0044	0.0015

information, and more informative prices can coexist with lower average posterior precision.

The China calibration illustrates how these mechanisms operate under partial access. Stock Connect gives foreign investors direct access to Type-I shares only, yet foreign entry changes the public price signal observed by domestic investors, who update beliefs, reallocate data-processing capacity, and adjust portfolios across both asset types. Partial access can therefore have market-wide and heterogeneous informational effects even on securities that remain inaccessible to foreign investors. The small scale of the active foreign sector is relevant because large informational effects can arise even when foreign ownership and participation are limited; neither is a sufficient statistic for informational influence. The transmission works through marginal informed demand and equilibrium prices.

These counterfactuals point to a broader lesson. The informational consequences of market access depend not only on which assets investors may hold, but also on whether they can translate private information into positions. In the China calibration, expanding foreign investability while retaining the short-sale constraint has little effect on price informativeness, whereas relax-

ing that constraint produces substantial improvements. Market access therefore reshapes informational efficiency through the joint determination of investability, information acquisition, and risk bearing.

Appendix: Proofs of Main Theoretical Results

This appendix supplies proofs of the paper's three main theoretical results. It covers the information-allocation proposition under heterogeneous market access (Proposition 1), the intermediate-precision cross-asset-learning proposition (Proposition 2), and the price-informativeness proposition (Proposition 3). Supporting lemmas, the constrained-equilibrium characterization, welfare derivations, and quantitative details appear in the Internet Appendix.

A.1. Proof of Proposition 1

Proof of Proposition 1. For domestic institutions, Lemma 1 reduces the data-choice problem to maximizing posterior precision subject to the binding capacity constraint. The interior first-order condition equalizes posterior precision in effective-data units, and the endpoint comparisons give (9).

For foreign institutions under partial access, Lemma 2 reduces the objective to Type-I Schur-complement precision. With $M \equiv M_{k,(\mathrm{II})}$ and $M_{k,(\mathrm{I})} = \bar{M} - M$, the marginal objective is $\frac{d\hat{\Omega}^{S}_{kyt,(\mathrm{I})}}{dM} = \Pi_{e,(\mathrm{I})}[\kappa(M) - 1]$. Holding the equilibrium price signal fixed, $\kappa(M)$ is continuous and, whenever the cross-precision is nonzero, strictly decreasing. The two endpoint conditions therefore give the Type-I and Type-II corners, while $\kappa(0) > 1 > \kappa(\bar{M})$ yields the unique interior solution $\kappa(M^{\text{int}}) = 1$. These are exactly the cases in (10). □

A.2. Proof of Proposition 2

Proof of Proposition 2. At the candidate all-Type-I foreign-processing corner, the foreign allocation rule is governed by the marginal spillover rate $\kappa_E(0)$. As $E \to 0$, the direct precision contribution of a Type-II observation vanishes, so $\kappa_E(0) \to 0$ and the Type-II allocation is zero. At the other extreme, equilibrium price precision rises sufficiently quickly with E that the cross-precision ratio z_E vanishes faster than the direct precision gain increases: $z_E^2 E \to 0$. Thus $\kappa_E(0) \to 0$ also as $E \to \infty$, and the Type-II allocation is again zero.

Internet Appendix Section IA.A.4.2 constructs a primitive lower bound for $\kappa_E(0)$ that is single-peaked in E. Under the stated sufficient condition, that lower bound exceeds one on a nonempty bounded interval. Hence $\kappa_E(0) > 1$ throughout that interval, the all-Type-I corner violates the foreign allocation rule, and strictly positive Type-II processing is optimal. The full bounds, high-precision case split, and lower-bound derivation are provided in that Internet Appendix section. □

A.3. Proof of Proposition 3

Proof. **Part (i).** The Type-II price-precision decomposition in Internet Appendix Section IA.D separates the change in the own-price component from the cross-price component available under partial access. Thus Type-II price informativeness falls if and only if

$$\left(\zeta^2_{(\mathrm{II});\mathrm{fu}} - \zeta^2_{(\mathrm{II})}\right)\Omega_{x,(\mathrm{II})} < \zeta^2_{(\mathrm{I}\leftarrow\mathrm{II})}\Omega_{x,(\mathrm{I})}.$$

Part (ii). At the all-Type-I foreign-processing corner, the Type-I SNR change has the same sign as the net change $\mathcal{H}_{\mathrm{I}}$ in aggregate Type-I hard-information precision. The posterior-precision accounting derived in the Internet Appendix is

$$\Delta\bar{\hat{\Omega}}_{y,(\mathrm{I})} = A_{\mathrm{I}}\mathcal{H}_{\mathrm{I}} - \frac{\lambda_K}{\lambda_2}C_{\mathrm{I}} - \mathcal{Q}_{\mathrm{I}},$$

where $A_{\mathrm{I}}, C_{\mathrm{I}} > 0$ and $\mathcal{Q}_{\mathrm{I}} \geq 0$ is the foreign portfolio-constraint discount. If Type-I price informativeness weakly falls, then $\mathcal{H}_{\mathrm{I}} \leq 0$, making the first and third terms nonpositive and the dilution term strictly negative. Average Type-I posterior precision therefore falls strictly. Conversely, when price informativeness rises, the joint outcome in the proposition occurs whenever

$$0 < \mathcal{H}_{\mathrm{I}} < \frac{(\lambda_K/\lambda_2)C_{\mathrm{I}} + \mathcal{Q}_{\mathrm{I}}}{A_{\mathrm{I}}}.$$

The left inequality makes Type-I price informativeness rise, while the right inequality makes average Type-I posterior precision fall. The full rearrangement is in the Internet Appendix. □

References

Bae, K.H., Bailey, W., Mao, C.X., 2006. Stock market liberalization and the information environment. *Journal of International Money and Finance* 25, 404–428.

Bekaert, G., Harvey, C.R., 2000. Foreign speculators and emerging equity markets. *The Journal of Finance* 55, 565–613.

Carpenter, J.N., Lu, F., Whitelaw, R.F., 2021. The real value of china ' s stock market. *Journal of Financial Economics* 139, 679–696.

Chan, M.K., Kwok, S., 2017. Risk-sharing, market imperfections, asset prices: Evidence from china ' s stock market liberalization. *Journal of Banking & Finance* 84, 166–187.

Chari, A., Henry, P.B., 2004. Risk sharing and asset prices: evidence from a natural experiment. *The Journal of Finance* 59, 1295–1324.

Choe, H., Kho, B.C., Stulz, R.M., 2005. Do domestic investors have an edge? the trading experience of foreign investors in korea. *The Review of Financial Studies* 18, 795–829.

Diamond, D.W., Verrecchia, R.E., 1981. Information aggregation in a noisy rational expectations economy. *Journal of Financial Economics* 9, 221–235.

Dvořák, T., 2005. Do domestic investors have an information advantage? evidence from indonesia. *The Journal of Finance* 60, 817–839.

Gao, C., Zeng, W., 2023. Near is dear: Remoteness, soft information, and stock returns. *SSRN Working Paper No. 4509746* URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4509746, doi:10.2139/ssrn.4509746. first version June 2022.

Gao, M., Huang, J., 2020. Informing the market: The effect of modern information technologies on information production. *The Review of Financial Studies* 33, 1367–1411.

Griffin, J.M., Kelly, P.J., Nardari, F., 2010. Do market efficiency measures yield correct inferences? a comparison of developed and emerging markets. *The Review of Financial Studies* 23, 3225–3277.

Grinblatt, M., Keloharju, M., 2000. The investment behavior and performance of various investor types: a study of finland's unique data set. *Journal of Financial Economics* 55, 43–67.

Grossman, S.J., Stiglitz, J.E., 1980. On the impossibility of informationally efficient markets. *The American Economic Review* 70, 393–408.

Hau, H., 2001. Location matters: An examination of trading profits. *The Journal of Finance* 56, 1959–1983.

Hellwig, M.F., 1980. On the aggregation of information in competitive markets. *Journal of Economic Theory* 22, 477–498.

Henry, P.B., 2000. Stock market liberalization, economic reform, and emerging market equity prices. *The Journal of Finance* 55, 529–564.

Henry, P.B., 2003. Capital-account liberalization, the cost of capital, and economic growth. *American Economic Review* 93, 91–96.

Kim, E.H., Singal, V., 2000. The fear of globalizing capital markets. *Emerging Markets Review* 1, 183–198.

Liberti, J.M., Petersen, M.A., 2019. Information: Hard and soft. *Review of Corporate Finance Studies* 8, 1–41.

Van Nieuwerburgh, S., Veldkamp, L., 2009. Information immobility and the home bias puzzle. *The Journal of Finance* 64, 1187–1215.

Verrecchia, R.E., 1982. Information acquisition in a noisy rational expectations economy. *Econometrica: Journal of the Econometric Society* 50, 1415–1430.